PRACTICAL NO 5

AIM:- Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network.

CODE

```
import tkinter as tk
from tkinter import messagebox


# ---------------- DIFFIE-HELLMAN FUNCTION ----------------

def calculate_keys(p, g, person_a_private, person_b_private):

    person_a_public = pow(g, person_a_private, p)
    person_b_public = pow(g, person_b_private, p)

    person_a_secret = pow(
        person_b_public,
        person_a_private,
        p
    )

    person_b_secret = pow(
        person_a_public,
        person_b_private,
        p
    )

    return (
        person_a_public,
        person_b_public,
        person_a_secret,
        person_b_secret
    )


# ---------------- CLI MODE ----------------

def cli_mode():

    while True:

        print("\n===================================")
        print("      DIFFIE-HELLMAN KEY EXCHANGE")
        print("===================================")
        print("1. Perform Key Exchange")
        print("2. Show Explanation")
        print("3. Exit")
```

```
choice = input("\nEnter your choice: ")

if choice == "1":

    try:
        p = int(input("\nEnter prime number (p): "))
        g = int(input("Enter primitive root (g): "))

        person_a_private = int(
            input("Enter Person A private key: ")
        )

        person_b_private = int(
            input("Enter Person B private key: ")
        )

        (
            person_a_public,
            person_b_public,
            person_a_secret,
            person_b_secret
        ) = calculate_keys(
            p,
            g,
            person_a_private,
            person_b_private
        )

        print("\n---------- RESULTS ----------")

        print(
            "Person A public key :",
            person_a_public
        )

        print(
            "Person B public key :",
            person_b_public
        )

        print(
            "Person A shared key :",
            person_a_secret
        )

        print(
            "Person B shared key :",
            person_b_secret
        )

        if person_a_secret == person_b_secret:
```

```
                print("\nKey exchange successful!")
                print(
                    "Both persons have the same shared secret."
                )
            else:
                print("\nKey exchange failed.")

        except ValueError:
            print("\nPlease enter numbers only.")

    elif choice == "2":

        print("\nDiffie-Hellman allows two persons to")
        print("create a common secret key over")
        print("an insecure network.")

        print("\nPrivate keys are never exchanged.")
        print("Only public keys are shared.")

    elif choice == "3":

        print("\nExiting program...")
        break

    else:

        print("\nInvalid choice. Try again.")


# ---------------- GUI FUNCTIONS ----------------

def gui_exchange():

    try:

        p = int(p_entry.get())
        g = int(g_entry.get())

        person_a_private = int(
            person_a_entry.get()
        )

        person_b_private = int(
            person_b_entry.get()
        )

        (
            person_a_public,
            person_b_public,
            person_a_secret,
            person_b_secret
```

```
        ) = calculate_keys(
            p,
            g,
            person_a_private,
            person_b_private
        )

        person_a_public_entry.delete(0, tk.END)
        person_a_public_entry.insert(
            0,
            str(person_a_public)
        )

        person_b_public_entry.delete(0, tk.END)
        person_b_public_entry.insert(
            0,
            str(person_b_public)
        )

        person_a_secret_entry.delete(0, tk.END)
        person_a_secret_entry.insert(
            0,
            str(person_a_secret)
        )

        person_b_secret_entry.delete(0, tk.END)
        person_b_secret_entry.insert(
            0,
            str(person_b_secret)
        )

        if person_a_secret == person_b_secret:

            result_label.config(
                text="✓ Key Exchange Successful\n"
                     "Both persons have the same shared secret."
            )

        else:

            result_label.config(
                text="✗ Key Exchange Failed"
            )

    except ValueError:

        messagebox.showerror(
            "Error",
            "Please enter valid numbers."
        )
```

```
# ---------------- CLEAR GUI ----------------

def clear_gui():

    entries = [
        p_entry,
        g_entry,
        person_a_entry,
        person_b_entry,
        person_a_public_entry,
        person_b_public_entry,
        person_a_secret_entry,
        person_b_secret_entry
    ]

    for entry in entries:
        entry.delete(0, tk.END)

    result_label.config(text="")


# ---------------- GUI MODE ----------------

def gui_mode():

    global p_entry
    global g_entry
    global person_a_entry
    global person_b_entry
    global person_a_public_entry
    global person_b_public_entry
    global person_a_secret_entry
    global person_b_secret_entry
    global result_label

    root = tk.Tk()

    root.title("Diffie-Hellman Key Exchange")
    root.geometry("600x600")

    tk.Label(
        root,
        text="DIFFIE-HELLMAN KEY EXCHANGE",
        font=("Arial", 18, "bold")
    ).pack(pady=15)

    tk.Label(
        root,
        text="Public Values",
        font=("Arial", 12, "bold")
```

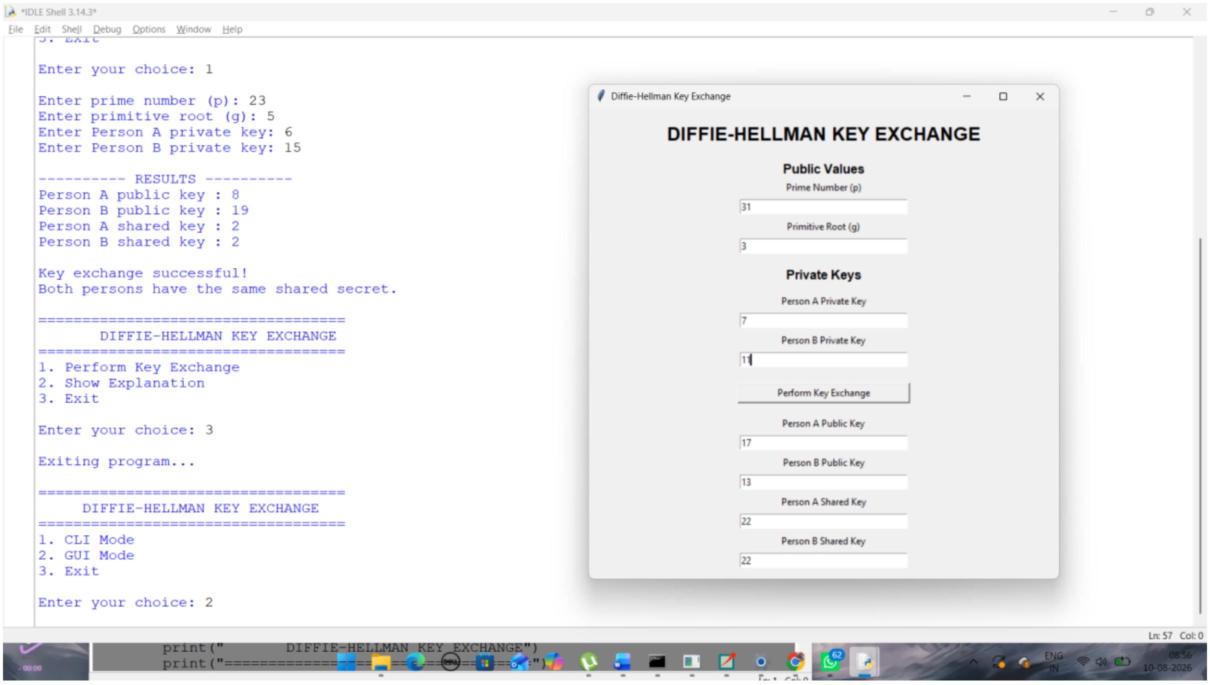
```
).pack()

tk.Label(
    root,
    text="Prime Number (p)"
).pack()

p_entry = tk.Entry(root, width=35)
p_entry.pack(pady=5)

tk.Label(
    root,
    text="Primitive Root (g)"
).pack()

g_entry = tk.Entry(root, width=35)
g_entry.pack(pady=5)

tk.Label(
    root,
    text="Private Keys",
    font=("Arial", 12, "bold")
).pack(pady=10)

tk.Label(
    root,
    text="Person A Private Key"
).pack()

person_a_entry = tk.Entry(root, width=35)
person_a_entry.pack(pady=5)

tk.Label(
    root,
    text="Person B Private Key"
).pack()

person_b_entry = tk.Entry(root, width=35)
person_b_entry.pack(pady=5)

tk.Button(
    root,
    text="Perform Key Exchange",
    command=gui_exchange,
    width=30
).pack(pady=15)

tk.Label(
    root,
    text="Person A Public Key"
).pack()
```

```
person_a_public_entry = tk.Entry(
    root,
    width=35
)
person_a_public_entry.pack(pady=5)

tk.Label(
    root,
    text="Person B Public Key"
).pack()

person_b_public_entry = tk.Entry(
    root,
    width=35
)
person_b_public_entry.pack(pady=5)

tk.Label(
    root,
    text="Person A Shared Key"
).pack()

person_a_secret_entry = tk.Entry(
    root,
    width=35
)
person_a_secret_entry.pack(pady=5)

tk.Label(
    root,
    text="Person B Shared Key"
).pack()

person_b_secret_entry = tk.Entry(
    root,
    width=35
)
person_b_secret_entry.pack(pady=5)

tk.Button(
    root,
    text="Clear",
    command=clear_gui,
    width=15
).pack(pady=10)

result_label = tk.Label(
    root,
    text="",
    font=("Arial", 11, "bold")
```

```
    )
    result_label.pack(pady=10)

    root.mainloop()


# ---------------- MAIN MENU ----------------

while True:

    print("\n==================================")
    print("    DIFFIE-HELLMAN KEY EXCHANGE")
    print("==================================")
    print("1. CLI Mode")
    print("2. GUI Mode")
    print("3. Exit")

    choice = input("\nEnter your choice: ")

    if choice == "1":

        cli_mode()

    elif choice == "2":

        gui_mode()

    elif choice == "3":

        print("\nProgram terminated.")
        break

    else:

        print("\nInvalid choice. Please try again.")
```

OUTPUT:-


IDLE Shell 3.14.3
File Edit Shell Debug Options Window Help
========== RESTART: C:/Users/as993/Downloads/T118 CS PRACTICAL NO 5.py =========
===================================
DIFFIE-HELLMAN KEY EXCHANGE
===================================
1. CLI Mode
2. GUI Mode
3. Exit
Enter your choice: 1
===================================
DIFFIE-HELLMAN KEY EXCHANGE
===================================
1. Perform Key Exchange
2. Show Explanation
3. Exit
Enter your choice: 1
Enter prime number (p): 23
Enter primitive root (g): 5
Enter Person A private key: 6
Enter Person B private key: 15
---------- RESULTS ----------
Person A public key : 8
Person B public key : 19
Person A shared key : 2
Person B shared key : 2
Key exchange successful!
Both persons have the same shared secret.
===================================
DIFFIE-HELLMAN KEY EXCHANGE
===================================
Ln: 45 Col: 19


IDLE Shell 3.14.3
File Edit Shell Debug Options Window Help
Enter your choice: 1
Enter prime number (p): 23
Enter primitive root (g): 5
Enter Person A private key: 6
Enter Person B private key: 15
---------- RESULTS ----------
Person A public key : 8
Person B public key : 19
Person A shared key : 2
Person B shared key : 2
Key exchange successful!
Both persons have the same shared secret.
===================================
DIFFIE-HELLMAN KEY EXCHANGE
===================================
1. Perform Key Exchange
2. Show Explanation
3. Exit
Enter your choice: 3
Exiting program...
===================================
DIFFIE-HELLMAN KEY EXCHANGE
===================================
1. CLI Mode
2. GUI Mode
3. Exit
Enter your choice: 2
Diffie-Hellman Key Exchange
DIFFIE-HELLMAN KEY EXCHANGE
Public Values
Prime Number (p)
31
Primitive Root (g)
3
Private Keys
Person A Private Key
7
Person B Private Key
11
Perform Key Exchange
Person A Public Key
17
Person B Public Key
13
Person A Shared Key
22
Person B Shared Key
22
Ln: 57 Col: 0